Claim Check · AI for Weather & Climate

For every claim decide: ✓ **Supported** (the figures support the claim) · ✗ **Refuted** (the figures contradict the claim) · ? **Cannot be determined** (our data cannot answer it). Name the figure(s) you use and give a one-sentence reason.

Group / names: _____

#	Claim	Verdict	Figure(s)	Reason
1	“GraphCast beats HRES for every variable and every lead time.”	☐ ✓ ☐ ✗ ☐ ?		
2	“The advantage of AI models over HRES grows with lead time.”	☐ ✓ ☐ ✗ ☐ ?		
3	“Physics-based models are more accurate in the tropics.”	☐ ✓ ☐ ✗ ☐ ?		
4	“The hybrid NeuralGCM is more accurate than the purely data-driven GraphCast.”	☐ ✓ ☐ ✗ ☐ ?		
5	“NeuralGCM predicts 2-m temperature better than HRES.”	☐ ✓ ☐ ✗ ☐ ?		
6	“AI models beat HRES in every region of the world.”	☐ ✓ ☐ ✗ ☐ ?		
7	“GenCast is the best ensemble for every variable.”	☐ ✓ ☐ ✗ ☐ ?		
8	“All ensembles are overconfident.”	☐ ✓ ☐ ✗ ☐ ?		
9	“Five GenCast members are better than 50 IFS ENS members.”	☐ ✓ ☐ ✗ ☐ ?		

#	Claim	Verdict	Figure(s)	Reason
10	“Therefore GenCast uses less energy than IFS ENS.”	☐ ✓ ☐ ✗ ☐ ?		
11	“Going from 20 to 50 members hardly improves the forecast.”	☐ ✓ ☐ ✗ ☐ ?		
12	“An ensemble mean is more accurate than any single forecast.”	☐ ✓ ☐ ✗ ☐ ?		
13	“Ensembles know where they will be wrong.”	☐ ✓ ☐ ✗ ☐ ?		
14	“The Hannover example shows that GenCast and IFS ENS are equally reliable.”	☐ ✓ ☐ ✗ ☐ ?		
15	“AI forecasts look as realistic as real weather maps.”	☐ ✓ ☐ ✗ ☐ ?		
16	“The forecast with the lowest error also gives the most realistic weather maps.”	☐ ✓ ☐ ✗ ☐ ?		
17	“All models get worse in unusual weather.”	☐ ✓ ☐ ✗ ☐ ?		
18	“AI models underestimate how strong extremes are.”	☐ ✓ ☐ ✗ ☐ ?		
19	“AI weather models will work just as well in the climate of 2050.”	☐ ✓ ☐ ✗ ☐ ?		
20	“AI weather models are more environmentally sustainable than physics-based models.”	☐ ✓ ☐ ✗ ☐ ?		